REAL Dimensional Consistency — Resolution (Working Draft v0.2)

Status tier: HYPOTHESIZED where noted (the dimensionless construction itself is a stated convention, not a discovered law). Structurally closed; numeric anchor values remain an empirical follow-on. No dependency on Pellis's simulation.

---

1. The direct check (and a correction to existing material)

REAL = `(E·M·R)/(T·S)` — Energy·Matter·Resonance over Time·Space.

A prior document in the corpus (Talonics, §4.4) asserts: *"Because REAL is a ratio of products, it is dimensionless."* This is checked directly and is **not correct as stated.** Using ordinary physical dimensions:

```
[E] = M·L²·T⁻²      [M] = M      [T] = T      [S] = L

(E·M)/(T·S)  =  (M·L²·T⁻² · M) / (T · L)  =  M²·L·T⁻³
```

That is not dimensionless. "A ratio of products" is not, by itself, a sufficient condition for dimensionlessness — only a ratio in which numerator and denominator dimensions actually cancel is. Pellis's original objection was correct, and the existing internal claim should be retired rather than carried forward into anything sent to him.

---

2. The actual fix: REAL is a constructed index, not a literal physical quantity

The resolution is not to find a hidden physical law that makes `(E·M·R)/(T·S)` cancel — it doesn't, and forcing one would be the kind of overclaim Pellis is positioned to catch. The honest fix is the same move used for standard dimensionless numbers in physics (Reynolds number, Mach number, etc.): **each term is a ratio to an explicit reference quantity of the same kind**, so each ratio is dimensionless by construction, and the product of dimensionless ratios is dimensionless.

```
REAL(t) := ( (E(t)/E_ref) · (M(t)/M_ref) · R(t) ) / ( (T(t)/T_ref) ·
(S(t)/S_ref) )
```

`R(t)` needs no reference scale — it is already dimensionless by construction from Prong 1 (`R(t) = σ(α(t)/α_ref)`, a bounded sigmoid output).

This is dimensionless by definition, not by discovered law. That distinction must be stated plainly in anything external: REAL's consistency is a modeling convention — a disciplined choice of how the four reference quantities are fixed — not a derived necessity. Claiming otherwise is exactly the overclaim risk flagged earlier in this work.

---

3. A path considered and retracted

An earlier working note proposed that `T_ref` and `S_ref` might not need inventing at all — that they could collapse directly onto Pellis's own renormalization dilation factors (`t → λᶻt`, `x → λx`), which would make the normalization "native" to his theory rather than an ERES-side patch.

This does not actually simplify things, and is retracted as the preferred path, for two reasons:

1. It would only buy invariance under Pellis's *specific* morphogenetic RG flow — not general dimensional consistency. REAL is used as an ERES governance index, not a claim about biological renormalization; tying it to his RG exponents would overstate what REAL is for.
2. "Matter" in REAL has no clean, named counterpart in Pellis's own variables (`ρ`, `P`, `g_ij`) to inherit a scaling dimension from. Forcing one would be invented, not native — the opposite of the intended simplification.

Conclusion: the reference-ratio normalization (§2) is the adopted resolution. This section is recorded so the retraction is visible, not silently dropped.

---

4. Defining the reference quantities — anchored to existing operational items, not invented fresh

Each reference quantity should be pinned to something already defined or already flagged as an open item in the corpus, rather than created new:

Term	Reference	Anchor
`T_ref`	characteristic timescale	the **window length** — already an open item in the BERA spec ("confirm operational default vs. the 30 s test fixture"). Resolving that item *is* defining `T_ref`; this should not be done twice, independently, in two documents.
`S_ref`	characteristic spatial scale	a baseline **VLSA deployment tier** (e.g. S0, the smallest/personal tier), per the existing tiered spatial framework.
`E_ref`	characteristic energy scale	**`C0`** — the baseline cross-channel coherence value already load-bearing in the binding `D(t) = [C0−C(t)]/C0` definition. No new quantity invented; reuses the one reference value already canonical in the corpus.
`M_ref`	characteristic mass/substrate scale	**GSSG mass at the personal-device tier** — the smallest unit in the existing GSSG deployment scale (personal device → building skin → hull), chosen for the same reason `S0` was chosen for `S_ref`: the baseline/smallest tier, not an arbitrary intermediate one.

Decision recorded (2026-06-19): both anchors adopted. Neither requires a new numeric value to be invented here — `C0` is already operationally defined (it must be measured to compute `D(t)` at all), and the GSSG personal-device-tier mass is an engineering quantity to be measured against existing GSSG specifications, not declared. What this document settles is *which* quantity plays the reference role, not its numeric value. That distinction should travel with the decision: REAL is now fully specified structurally; its numbers are an empirical follow-on, not a derivation owed to anyone.

One dependency carried forward deliberately, not by oversight: `E_ref = C0` means REAL inherits whatever the window-length resolution (open item, §4 `T_ref` row) settles on, since `C0` is itself computed over a window. Resolving window length closes both `T_ref` and, downstream, the operational meaning of `E_ref` in one move — this is a feature of anchoring to existing infrastructure rather than inventing parallel definitions, not a new problem.

---

5. What invariance actually means here

Once references are fixed, REAL is invariant under any global rescaling of units **because** numerator and denominator co-scale by definition — e.g., doubling the unit used to measure `E` doubles both `E` and `E_ref` identically, leaving `E/E_ref` unchanged. This is true by construction, not a theorem requiring proof. It should be presented to Pellis exactly that way: a stated convention with a one-line justification, not a result claimed to follow from deeper necessity.

---

6. Status

Resolved, no dependency on Pellis:

- The original "automatically dimensionless" claim is corrected
- The reference-ratio construction is adopted and stated honestly as convention, not law
- The RG-dilation alternative is considered and explicitly retracted, with reasons recorded
- All four reference quantities are now anchored to existing corpus infrastructure: `T_ref` and `E_ref` both resolve through the window-length item; `S_ref` and `M_ref` both resolve through baseline/smallest-tier choices in their respective existing scales (VLSA, GSSG)

Genuinely open, by design — empirical, not definitional:

- The window-length value itself (already on the v2 audit list) — this document does not duplicate that decision, it inherits it
- The actual measured numeric values of `C0` and the GSSG personal-device-tier mass — these are measurements to be taken, not numbers to be assumed for this document to be considered structurally closed

This document is structurally closed as of this revision. What remains is empirical (measuring the anchors), not conceptual (deciding what the anchors are).